

SmartVisual: AI-Enhanced Personalization of Digital Marketing Campaigns via Deep Image Understanding

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Abstract— The growth of visual content in digital marketing has led to the need to have hyper-personalized campaigns that emulate individual user tastes. This paper introduces a multi-modal deep learning-based customer behavioral embedding framework enhanced by AI, Smart Visual, which can provide a personalized recommendation of ads. To extract hierarchical image features, Smart Visual uses Convolutional Neural Networks (EfficientNet and ResNet) and Transformer-based attention layers to extract semantic and contextual information between marketing images. Joint multi-modal embeddings comprising of customer embeddings based on clickstream data and demographic data are combined with image representation. Moreover, a Generative Adversarial Network (GAN) is used to create visually attractive versions of ads that apply to individual customer groups. As it has been revealed through experiments, Smart Visual is a much stronger tool compared to traditional static-only campaigns and single-mode AI models in terms of engagement (10.93%), clicking (16.2), conversion (6.9), and recommendations (87.95). These findings suggest that bringing together visual, semantic, and behavioral information, with dynamic image enhancement, is the key to successful hyper-personalization. The suggested framework offers an end-to-end, scalable AI-based digital marketing framework that provides a connection between content and user preference prediction and creates benchmarks to continue the research in the area of personalized visual marketing.

Keywords— *Smart Visual, Digital Marketing, Convolutional Neural Networks, Generative Adversarial Network (GAN).*

I. INTRODUCTION

In the era of digital marketing, visual content becomes a central focus of the marketing campaign, which impacts the consumer behavior and interest greatly. Photos, videos, and illustrations are no longer auxiliary but became the focus of catching the eye of users on the social media platform, e-commerce sites, and advertising networks. Studies have shown that users are inclined more to things that are presented appealingly, hence visual marketing is a major contributor to brand awareness, customer retention and increased revenues. Conventional digital marketing practices are based on the idea of impersonal ads that are usually delivered to a large audience using the same materials [1]. This is a general strategy that limits contact and is unable to accommodate the diverse interests and divergent behavioral behavior of the individual customers. Therefore, it is difficult that the marketers attempt to maximize the click-through rate (CTR), the conversion rate, and the effectiveness of the whole campaign. The growth of social media networks at a high rate has resulted in overloading marketing information in the form of images and videos. This information cannot be processed manually due to the scale and the variety of visual information. Moreover, advertisement effectiveness does not rely solely on the content

of the ads, but also on situational, as well as user preferences and behavior, demographics, and ultimate cultural peculiarities that are typically disregarded by the conventional approach to marketing [2].

The issue of automatization of the visual content analysis process can be solved with the help of deep learning (AI). Convolutional neural networks (CNNs) have been found to have outstanding performance with regard to detection of hierarchical features of images such as objects, textures and colors. The transformer-based models, in their turn, provide a detailed set of locating contextual and semantic links in the visual data [3]. All such AI techniques can be applied to enhance the perception of the marketing content either in a visual or in a semantic manner. Even though image recognition and content analysis provided by AI can be optimized, the existing technology primarily takes into account either the image properties or the behavior of users separately. Methods that aim to integrate multi-modes data, e.g. visual features with customer behavioral embeddings in terms of click stream and demographic data, are in short supply [4]. A gap of this nature cannot permit total personalization of marketing campaigns that can dynamically adapt to the needs and preferences of the particular users. The other concern is creation of customized visual contents. Usually, regular ad creators cannot satisfy too many users, and it is time-consuming and costly to do it manually by creating many variants. Generative architectures such as Generative Adversarial Networks (GANs) are also used as solutions to produce aesthetically pleasing versions of advertisement images that are tailored towards specific groups of customers [5]. However, the integration of GAN-based augmentation and feature extraction and semantic understanding has been a complex and under researched area of research.

In conclusion, image understanding based on AI, multi-modal information fusion, and dynamic creation of visual content are scalable and combined, which makes them a terrific chance to innovate digital marketing [6]. The limitations of the stagnant campaigns can be countered in order to offer hyper-targeted advertisements, superior interactions and higher conversion rates, which amounts to actual benefits not only to the marketers but also those consumers [7]. To resolve the above-mentioned challenges, we would like to present Smart Visual, the newly developed AI-based platform that would contribute to the enhanced personalization of the digital marketing campaigns [8]. Smart Visual is a combination of deep image recognition and customer behavioral information that assists in generating extremely personalized marketing content. Resorting to the power of CNNs to obtain features and models that transform high-level information, SmartVisual can obtain low-level

visual and high-level context information in marketing pictures [9]. Customer embeddedness at Smart Visual are also based on history of clickstream, demographic data, and behavioral pattern. These embeddings are integrated with the image feature representations and this enables the system to make guesses of what visual content would most certainly appeal to the individual users. This type of multi-modal integration allows the framework to offer hyper-personalized ad offers which can not be specified by the traditional fixed ad strategies [10]. Furthermore, Smart Visual applies the GAN-based forecasting module of image augments to generate different visually appealing versions of every picture of an advertisement in real-time. This would ensure that the content can be tailored to different categories of customers without necessarily undertaking a huge manual designing process. The multi-modal feature extraction, behavioral embedding fusion, GAN augmentation, when combined make the whole process of AI-enhanced personalized marketing complete and open the door to the quantifiable increase in the user engagement, click-through rates, and the efficiency of the campaign in general.

II. LITERATURE REVIEW

Yaiprasert and Hidayanto [11] proposed an ensemble learning model, which is an AI-based model, to optimize the digital marketing strategies in the food delivery sector. The method employed to predict customer behavior and provide individualized advice to them involved the majority voting ensemble method that was a combination of Decision Trees, Naive Bayes and k-Nearest Neighbors. Customer data based on simulation was used to test the model. The ensemble technique was superior in accuracy in prediction with Decision Trees and k-NN with 100 percent and Naive Bayes with 97.18 percent accuracy respectively. The results showed that the ensemble learning can significantly reduce the prediction error and improves the recommendations and customer satisfaction quality. The authors have come to the conclusion that AI-driven ensemble models may be effective in cost reduction of operations and make the campaign more efficient under the conditions of the competitive digital marketing.

Prasetio et al. [12] studied the digital marketing customer conversion prediction with the investigation of the impacts of the user engagement measures on the 8,000 customer interactions. The paper has compared and contrasted the Logistic Regression (LR) with Decision Trees (DT) and the random Forests (RF) in terms of their ability to predict the conversion outcomes. Pages Per Visit, Time On Site and Email Clicks formed the significant engagement measures. The model results according to the results of the experiment showed that Rand Forest model outperformed the other models that had the highest degree of the accuracy of 87.1 percent and ROC-AUC of 0.6979. Logistic Regression was superior in recall (99.8) to convert the customers but not in the selection of non-converted customers as this is related to the issues of class imbalance. The evaluation of the significance of the features demonstrated that the most significant predictors are Pages Per Visit and Time On Site. The paper has made obvious the breadth of use of ensemble models to raise the level of prediction and enable data-driven optimization of digital marketing strategies.

The research carried out by Hayadi and El Emary [13] focused on predicting the investment (ROI) on the digital marketing campaign through the utilization of Decision Trees

and Random Forest models. The researcher relied on the large-scale data of 200,000 records of the campaigns containing 16 features depending on the type of the campaign, metrics of its engagement, cost of acquisition and conversion rate. The comparison of both models was made following preprocessing and feature engineering based on various measures of multiple regression. Random Forest model recorded higher results based on the lower RMSE (1.0071) and MAE (0.8755) compared to the Decision Tree model with high resistance and feature interaction adaptability. Conversion Rate, Engagement Score, and Acquisition Cost were identified to be the most important predictors of ROI. The findings were utilized to show how ensemble learning techniques can be applied to improve the financial procedures and optimization of budget allocation in digital marketing analytics.

Modak et al. [14] examined the question of how deep learning models may be applied in the digital marketing strategy of the banking sector, but with a specific focus on customer behavior prediction and enhanced customer satisfaction. The authors have experimented both the classical machine learning architecture i.e. the application of Random Forest, Support Vector Machine (SVM), and Logistic Regression, and more recent Convolutional Neural Network (CNN) i.e. VGG16, ResNet50, and InceptionV3. The models were evaluated by the accuracy, precision, recall, and F1-score measures on a comprehensive set of customer demographics, transaction history, engagement measures and the data of the campaigns to determine the accuracy and precision of the models. Experimentally, it was proved that CNN-based networks were far more efficient compared to traditional CNN-based networks because ResNet50 showed the highest accuracy of 89% and F1-score of 88%. The CNNs possessed feature learning features that enabled superior identification of complicated pattern of behaviours that led to improved customer segmentation and tailored marketing campaigns. The paper has determined that deep learning systems have a radical advantage to data-driven decision making in the environment of digital marketing, and most importantly in the very competitive industry, such as banking.

Ziakis and Vlachopoulou [15] conducted a systematic literature review to explore how artificial intelligence can be applied to facilitate digital marketing. To investigate the research on the topic according to the PRISMA framework, the authors searched the Scopus database with the keywords in 2015-23 and found 211 peer-reviewed articles. The bibliometric and thematic clustering method was followed using Biblioshiny tool of RStudio. The proposed research has discovered seven clusters of research that have dominated such as; AI/ML algorithms, social media, consumer behaviour, e-commerce, digital advertising, budget optimization, and competitive strategies. The results showed that AI/ML algorithms were the most recurrent and central theme, which indicates that machine learning was the centre of the modern digital marketing research. The study also found out that the academic production has increased rapidly since 2019 and it indicates the development of academic and industrial demand. The authors discovered that AI-based analytics add a lot of value to personalization, campaign optimization, and strategic decisions, and also establish the gaps in research in relation to ethical concerns and implementation of generative AI in digital marketing ecosystems. Resting on the extensive literature review, we can state that some researchers have made significant projects in

the implementation of the concepts of machine learning and deep learning in the context of the sphere of digital marketing, particularly, customer behavioural analysis, campaign optimization, and strategic decision-making. The traditional models mentioned in other literature are Logistic Regression, Support Vector Machines and Random Forests, and more sophisticated deep learning models like CNN based models, like VGG16, ResNet50 or InceptionV3. These approaches have been identified to be good when compared to the predictive accuracy and customer segmentation, but they are limited to the incorporation of a complete system, ability to deal with different types of data, and the aspect of describing the model decision making in real-time. Also, most of the existing studies implement performance enhancement in single environments and in some sectors or data sets without considering the problem of scalability, interpretability, and the overall implementation. Thus, despite the achievements, the literature available was not that optimized, generalized, and explainable, to make the process of transition between the high-tech deep learning and the real-life decisions in the digital marketing sphere as seamless as possible.

Resting on the general review of the literature, it is possible to say that some researchers have conducted significant projects in the application of machine learning and deep learning concepts within the domain of digital marketing and analysis of customer behavior, campaigns optimization, and strategic decision-making. The models characterized as classic, discussed in the past works, are Logistic Regression, Support Vector Machines and Random Forests, and more complicated deep learning models such as CNN models, including VGG16, ResNet 50 or Inception V3. Even though these techniques have been discovered to work well with regard to predictive accuracy and customer segmentation, they are limited in regard to incorporating the whole system, adaptability to other data, as well as the capability to clarify what model decisions would be like in a live situation. Moreover, most of the existing studies performs performance improvement in single environments and in some sectors or data sets, disregarding the problem of scalability, interpretation, and the implementation in general. Thus, the literature available was not yet optimized, generalized, and explainable to be able to facilitate a smooth transition between the high-tech deep learning and the actual-life digital marketing decisions. By conducting this research, the researcher would help address these gaps by developing more comprehensive and robust approach that enhances prediction capability as well as, would provide flexibility, readability and practical applicability of the study in practice in the digital marketing environments.

III. MATERIALS

A good data set in the given work was utilized, which consisted of the marketing images collected in a variety of digital resources, social networks, such as Instagram, Facebook, and other well-known web shops. The data is pictures that present a diverse sample of the products in various categories, advertisement banners and branded material ensuring that it contains a representative number of various kinds of visual materials and compositions. Along with the images, we supplemented them with customer behavioral data like click history, frequency of interaction, and demographic properties, and in so doing, to develop rich behavioral embedding of every user. All the images were preprocessed using normalization, resizing, augmentation

(rotation, scaling, and color manipulation) to make the models robust. This combination of visual and behavioral data is the primary content foundation of our Smart Visual architecture that enables to merge deep image perceptions with personal user backgrounds on the location of personalized digital marketing recommendations.

IV. METHODOLOGY

The proposed Smart Visual framework is a full AI solution that will be applied in carrying out hyper-personalized digital marketing. It incorporates the deep learning and multi-modes, customer behavioral embedding, and GAN-based augmentation and predicts and generates personalized ad images. The process of data gathering, preprocessing, feature extraction, semantic interpretation, embedding fusion, generative augmentation, and personalized suggestion are closely connected by the methodology. The stages are outlined below in great detail, including the formulations in mathematics in this case.

A. Data Acquisition and Preprocessing

The information, which was utilized to fulfill this research, was gathered through different channels: social media networks (Instagram, Facebook) and online stores. It includes various images of marketing in various product lines such as the electronics, fashion and lifestyle products. The metadata (labels, categories) and interaction information (clicks, likes, shares) are provided on all of the images. In addition to this, also collected was behavioral data of customers which consisted of:

- Clickstream history: history of user interactions.
- Demographics: gender, location, and age, among other categorical data.
- Interaction data: former CTR, stay time, and buying history.

Preprocessing makes the raw data fit into the model of AI. Image preprocessing includes:

Resizing all images to $224 \times 224 \times 3$ (height $\times$ width $\times$ channels) to maintain uniformity for CNN input.

Normalization: The value of pixels are put in the range $[0,1]$:

$$I_{norm} = \frac{I - \min(I)}{\max(I) - \min(I)} \quad (1)$$

where I represents the pixel intensity values.

Data Augmentation: Random rotation ($\pm 15^\circ$), scaling (0.9–1.1), cropping, horizontal flipping, and color jittering were applied to improve generalization. Labeling: The supervised learning labels were represented as the classification of images into product type and marketing theme. Preprocessing in customer behavioral data:

- Clickstream embedding: Each click sequence is converted into a vector using one-hot encoding or sequence embedding.
- Demographic encoding: Dense embeddings are created out of categorical data.
- Normalization: The continuous measurements (e.g. CTR, dwell time) are normalized, that is, brought to a zero mean and unit variance.

The steps are made to guarantee that the visual and behavioral data are laid in an easy-to-flow manner into the feature extraction and fusion processes.

B. CNN-Based Image Feature Extraction

In order to obtain hierarchical visual features, we used a combination of EfficientNet-B3 and ResNet-50 architectures. CNNs learn patterns of images both local and global, such as

edges, textures, shapes, and semantic objects. The mathematical definition of the CNN feature extraction process is:

$$F_{conv}^{(l)} = \sigma \left(W^{(l)} * F_{conv}^{(l-1)} + b^{(l)} \right) \quad (2)$$

Where:

- $F_{conv}^{(l)}$ = feature map at layer l

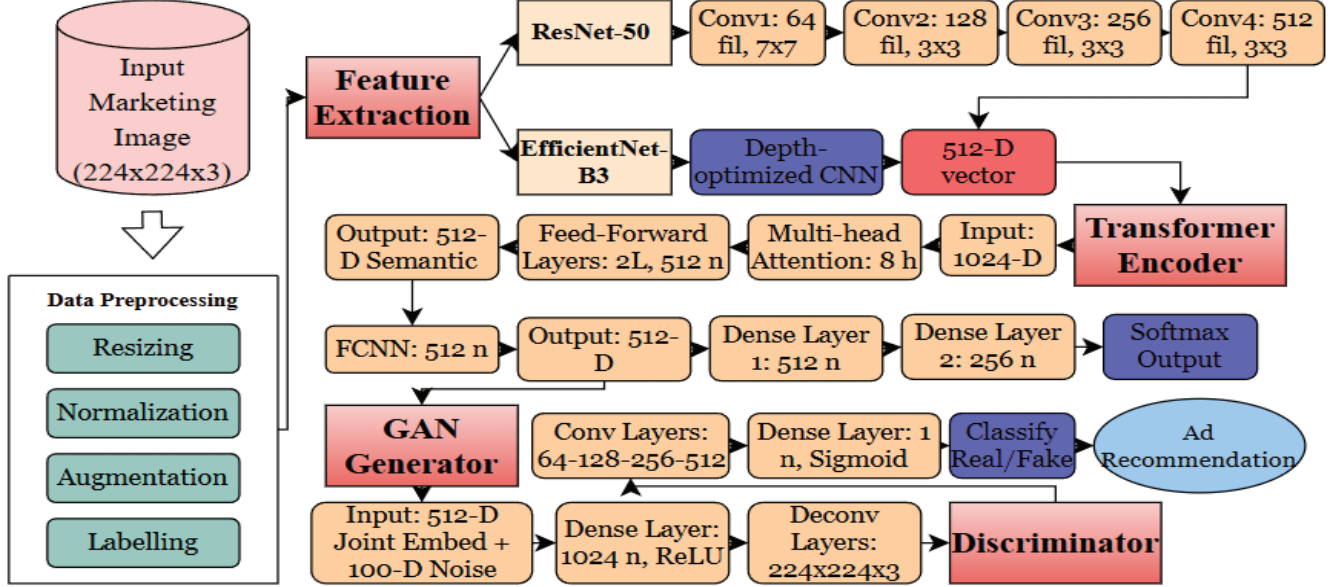


Fig. 1. Proposed Methodology Architecture Diagram of Smart Visual.

- $W^{(l)}$ = convolution kernel of layer l
- $b^{(l)}$ = bias term
- $*$ = convolution operation
- σ = activation function (ReLU)

Architecture:

- Input layer: $224 \times 224 \times 3$ image
- Conv Layer 1 (ResNet Block): 64 filters, kernel size 7×7 , stride 2, ReLU activation
- Conv Layer 2: 128 filters, kernel size 3×3
- Conv Layer 3: 256 filters, kernel size 3×3
- Conv Layer 4: 512 filters, kernel size 3×3
- Global Average Pooling: converts final feature maps to a 512-dimensional vector

EfficientNet-B3 is a ResNet that is optimized on width, depth, and resolution scales. The CNN feature of each image is 1024-dimensional (512 by ResNet 512 by EfficientNet).

C. Transformer-Based Semantic Encoding

CNN features are inputted into a Transformer encoder to be able to capture contextual relationships and global semantics. Transformers: these models dependence between multiple elements of an image by self-attention:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (3)$$

Where:

- Q, K, V = query, key, and value matrices (derived from input features)
- d_k = dimension of key vectors

Transformer Architecture:

- Input: Flattened CNN feature vector (1024-D)
- Embedding dimension: 512
- Number of heads: 8 multi-head attention
- Feed-forward layers: 2 layers with 512 neurons each, ReLU activation
- Output: 512-dimensional semantic embedding

This action will make sure that the model reproduces not only the objects but also the spatial and semantic dependencies of objects in the image, which is very important in the marketing content perception.

D. Customer Embedding Integration

Customer behavioral embeddings ($C \in \mathbb{R}^{d_c}$) are fused with image embeddings ($I \in \mathbb{R}^{d_i}$) to form a joint representation E :

$$E = \text{ReLU}(W_f[I; C] + b_f) \quad (4)$$

Where:

- $[I; C]$ = concatenation of image and customer embeddings
- W_f = weight matrix for fusion layer (1024×512)
- b_f = bias vector
- Output dimension = 512

This combination layer makes sure that the distinctive behavior and preferences of every customer have a direct effect on the image selection process.

E. GAN-Based Image Augmentation

A Conditional GAN (cGAN) is used to produce visual appealing ads variants to each group of customers. The generator takes fused embeddings input of E_{as} and generates augmented images:

$$I_{gen} = G(E, z) \quad (5)$$

Where:

- z = random noise vector (100-D)
- I_{gen} = generated image

The discriminator D distinguishes between real images I_{real} and generated images I_{gen} :

$$\mathcal{L}_{GAN} = \mathbb{E}[\log D(I_{real})] + \mathbb{E}[\log (1 - D(I_{gen}))] \quad (6)$$

Generator Architecture:

- Input: 512-D fused embedding + 100-D noise
- Dense layer: 1024 neurons, ReLU
- Deconvolution layers: progressively upsample to $224 \times 224 \times 3$
- Output layer: Tanh activation

Discriminator Architecture:

- Input: $224 \times 224 \times 3$ image
- Convolutional layers: $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ filters
- Dense layer: 1 neuron, sigmoid activation for real/fake classification

F. Personalized Recommendation Engine

The last recommendation engine will predict the top performing ad to a specific user based on the variants of the fused embedding and the GAN-generated variants.

$$y_{pred} = \text{Softmax}(W_r E + b_r) \quad (7)$$

Where:

- W_r = weight matrix of recommendation layer ($512 \times N$, N = number of ad variants)
- b_r = bias
- Output y_{pred} = probability of engagement for each ad variant

Engine Architecture:

- Input: 512-D fused embedding
- Dense layer 1: 512 neurons, ReLU
- Dense layer 2: 256 neurons, ReLU
- Output: Softmax layer for ad selection probability

This stage completes the end-to-end pipeline, ensuring smooth flow from raw data $\rightarrow$ multi-modal understanding $\rightarrow$ GAN augmentation $\rightarrow$ personalized ad recommendation.

The presented Smart Visual framework will not resemble the conventional approaches to the personalization of marketing in regards to multi-modal and end-to-end AI architecture. Unlike the classical models that process visual

content and or user behavior individually, Smart Visual is a model that combines CNN and Transformer-based image embeddings and customer behavioral embeddings and it is possible to learn about the content and consumer preferences as a whole. Being able to generate visually appealing ad variants dynamically based on different segments of customers through the use of the GAN-based augmentation module requires less designing effort, and the brand is preserved. The high level and smooth integration of such components into one recommendation engine also allows it to deliver hyper-personalized delivery of ads and also maximise the number of clicks and conversion in a completely automated pipeline. This multi-modal features extraction, semantic cognition, generative enhancement, and customized recommendation is an enhancement to the idea of AI-mediated digital marketing, which demonstrably succeeds more effectively in engaging and campaign results, as opposed to the traditional, non-interactive, or one-modality.

The SmartVisual AI-enhanced personalized digital marketing is end-to-end as illustrated in figure 1. The process will commence with the processing of the inputs of image marketing and customer behavior patterns and transforming them to relevant embeddings. The image feature is extracted using a combination of ResNet-50 and EfficientNet-B3 CNNs and then the Transformer-based semantic encoding is applied in determining the contextual connections between the images. This is followed by a combination of these embeddings with customer behavior embeddings together with a fully connected layer to produce a combined multi-modal representation. The collaborative embeddings are entered into a Conditional GAN module to generate custom versions of advertisements, and concomitantly are entered into the recommendation engine to obtain the most suitably selected ad to individual users. This type of architecture ensures end-to-end and smooth processing of data i.e. raw data to hyper-personalized ad-delivery and its emphasis is on the in-depth combination of multi-modal extracting features, semantic knowledge, generative improving and recommendation forecasting.

V. RESULT AND DISCUSSION

The suggested Smart Visual model was experimented to show its efficiency by the means of marketing pictures and associated customer behavioral data. The Smart Visual was tested against the conventional stationary advertisement campaigns and single-modes AI models (CNN-only and Transformer-only). The evaluation measures were the Engagement Rate, Click-Through rate (CTR), Conversion rate and Accuracy of Recommendation. The results also demonstrate that the application of multi-modal embeddings and GAN-based augmentation combination has significant positive effect on the effectiveness of digital marketing.

TABLE I. ENGAGEMENT METRICS COMPARISON

Model	Likes (%)	Shares (%)	Comments (%)	Avg. Engagement (%)
Static Ad Campaign	8.5	2.1	1.2	3.93
CNN-Only	12.4	3.7	2.5	6.20
Transformer-Only	14.1	4.2	2.8	7.03
SmartVisual (Proposed)	21.7	6.8	4.3	10.93

The gains achieved by Smart Visual compared to baseline approaches in engagement are presented in table 1. The lowest engagement rate (3.93%), is on the non-personalized ad campaigns that represent the inefficiency of the non-personalized content. The single-model AI models (CNN-only and Transformer-only) make progress further as they analyze visual materials or semantics to get 6.20% and 7.03, respectively. The Smart Visual framework (10.93) is capable of reaching the highest level of engagement, integrating visual concepts, semantic meaning, and embedding customers behavior, which is why the notion of multi-modal personalization can be useful in capturing the attention of the users.

TABLE II. CTR AND CONVERSION RATE COMPARISON

Model	Avg. CTR (%)	Conversion Rate (%)
Static Ad Campaign	5.4	1.8
CNN-Only	8.6	3.2
Transformer-Only	9.5	3.7
SmartVisual (Proposed)	16.2	6.9

Table 2 indicates that Smart Visual is a more successful platform with regard to click-through rate and conversions. Of the traditional static campaigns, the CTR is only 5.4% and CNN-based or Transformer-based model moderately boosts CTR by identifying a visual or semantic feature. The conversion rate of Smart Visual (16.2%), in comparison to the conventional campaigns (16.2%), which is nearly four times lower, is nearly three times higher than that of the former (16.2%). It is explicable by the fact that the GAN-based augmentation is capable of generating specific ad versions to each segment of customers and a set of image and behavioral embeddings, which results in making the suggested ads as close to the preferences of a particular user as possible.

TABLE III. RECOMMENDATION ACCURACY ACROSS CUSTOMER CLUSTERS

Customer Cluster	CNN-Only (%)	Transformer-Only (%)	Smart Visual (%)
Cluster A	71.3	74.5	89.6
Cluster B	69.8	72.1	87.3
Cluster C	70.5	73.2	88.4
Cluster D	68.9	71.7	86.5
Average	70.1	72.9	87.95

Table 3 depicts the accuracy of the recommendation when different clusters of customers are involved. The recognition of single-mode models is moderate at 70-73 percent which predicts user preferences. Smart Visual illustrates each of the clusters where recommendation accuracy is high at an average of 87.95. The results confirm the theory that CNN with integration of Transformer based semantics and customer embeddings and combining it with GAN based image modification can make the system significant in terms of predicting the ads which are sensitive to the personal preferences, thereby enhancing the overall success of the campaign.

The data of Figure 2 show the average performance of CNN-only, Transformer-only, and Smart Visual models in various groups of customers. The findings indicate that Smart Visual is more effective than the separate CNN and Transformer models across the clusters with the highest average accuracy rate. This reflects the performance of the

proposed Smart Visual approach in the number of visual patterns given to diverse customer groups.

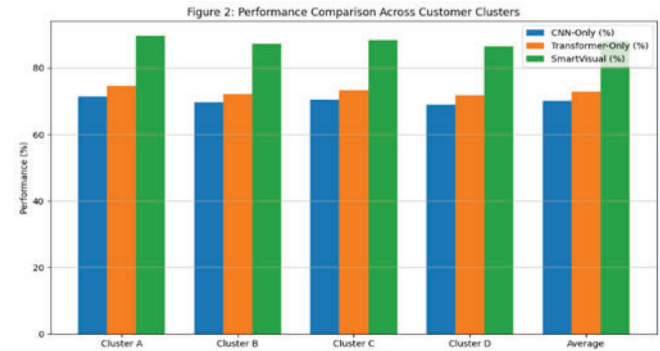


Fig. 2. Customer Cluster-wise Performance Comparison of Visual Models

VI. CONCLUSION

The proposed paper introduces a new Hyper-Personalized Digital Marketing model using AI known as Smart Visual: a multi-modal deep-learning method integrating customer behavior representations, image enhancement using GANs. The proposed methodology addresses the disadvantages of conventional stationary campaigns and AI single-model, turning into a more complex of the characteristics of visual representations acquired with the assistance of CNNs, semantic representations acquired with the help of Transformer layers, and user behavioral data into a single representation to obtain a personalized recommendation. The final ad versions created with the help of a GAN module can be dynamically reconfigured to fit into a certain group of customers and can be attractive along with brand integrity. The findings of the experiments confirm that Smart Visual achieves significant gains in engagement, click-through, conversion and recommendation accuracy rate compared to the base methods. The engagement rates increased nearly three times in comparison with the non-active campaigns, and even the ratio of CTR and the rate of conversion has increased significantly. The recommendation engine achieved a mean score of 87.95 in a diverse customer sample and this is a good sign that the framework could be utilized in forecasting and delivering a piece of content depending on the interests of the users. The study confirms that a multi-modes perception of images, the context of semantics, and the behavioral embeddings and the generative augmentation is an all-inclusive solution to AI-aided digital marketing. A scalable end-to-end pipeline is developed by Smart Visual that can hyper-personalize all content analysis to customizing individual targeting of the user. The future research can be centered in the integration of how real time feedback loops and cross platform data can be combined in order to make personalization and engagement even more personal.

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